Introduction to Artificial intelligence
Tutorial Sheet 1 (correction)

1.1 Define in your own words: (a) intelligence, (b) artificial intelligence, (c) agent, (d) rationality, (e) logical reasoning.

a. Dictionary definitions of intelligence talk about "the capacity to acquire and apply knowledge" or "the faculty of thought and reason" or "the ability to comprehend and profit from experience." These are all reasonable answers, but if we want something quantifiable we would use something like "the ability to act successfully across a wide range of objectives in complex environments."

b. We define artificial intelligence as the study and construction of agent programs that perform well in a given class of environments, for a given agent architecture; they do the right thing. An important part of that is dealing with the uncertainty of what the current state is, what the outcome of possible actions might be, and what is it that we really desire.

c. We define an agent as an entity that takes action in response to percepts from an environment.

d. We define rationality as the property of a system which does the "right thing" given what it knows. See Section 2.2 for a more complete discussion. The basic concept is perfect rationality; Section ?? describes the impossibility of achieving perfect rationality and proposes an alternative definition.

e. We define logical reasoning as the a process of deriving new sentences from old, such that the new sentences are necessarily true if the old ones are true. (Notice that does not refer to any specific syntax or formal language, but it does require a well-defined notion of truth.)

1.2 Are reflex actions (such as flinching from a hot stove) rational? Are they intelligent?

Yes, they are rational, because slower, deliberative actions would tend to result in more damage to the hand. If "intelligent" means "applying knowledge" or "using thought and reasoning" then it does not require intelligence to make a reflex action

1.3 The Turing Test[1] is a deceptively simple method of determining whether a machine can demonstrate human intelligence: If a machine can engage in a conversation with a human without being detected as a machine, it has demonstrated human intelligence. The test is conducted in an interrogation room run by a judge. The test subjects, a person and a computer program, are hidden from view. The judge has a conversation with both parties and attempts to identify which is the human and which is the computer, based on the quality of their conversation. Turing concludes that if the judge can't tell the difference, the computer has succeeded in demonstrating human intelligence. That is, it can think. He predicts that, by the year 2000, a computer will have a 30% chance of passing a five-minute Turing Test with an unskilled interrogator. What chance do you think a computer would have today?

Today, artificial intelligence has advanced significantly since Turing's prediction. Large language models (LLMs) like ChatGPT, GPT-4, and other AI-powered chatbots have become highly sophisticated in generating human-like responses. Many modern AI systems

[1] Read Turing's original paper on AI (Turing, 1950): https://academic.oup.com/mind/article/LIX/236/433/986238

can already pass a basic Turing Test in short conversations, especially with unskilled interrogators.
In controlled Turing Test experiments, AI models have been shown to fool human judges at rates exceeding Turing's predicted 30% threshold. Some chatbots, such as Google's LaMDA, OpenAI's ChatGPT, and Meta's Llama, can generate coherent and contextually relevant responses that often appear indistinguishable from human replies. However, in extended conversations or under scrutiny by skilled interrogators, these models still exhibit limitations—such as inconsistencies, a lack of true understanding, and occasional nonsensical or incorrect answers.
Given these advancements, it's reasonable to estimate that a modern AI system could have a **much higher than 30% chance** of passing a five-minute Turing Test, likely exceeding **50% or more**, depending on the context and the skill level of the judge. In casual conversations, AI could even pass the test the majority of the time. However, true human-like intelligence—complete with reasoning, common sense, and deep understanding—remains an ongoing challenge in AI research.

1.4 The neural structure of the sea slug *Aplysia* has been widely studied (first by Nobel Laureate Eric Kandel) because it has only about 20,000 neurons, most of them large and easily manipulated. Assuming that the cycle time for an *Aplysia* neuron is roughly the same as for a human neuron, how does the computational power, in terms of memory updates per second, compare with the high-end computer?

	Supercomputer	Personal Computer	Human Brain
Computational units	10^4 CPUs, 10^{12} transistors	4 CPUs, 10^9 transistors	10^{11} neurons
Storage units	10^{14} bits RAM	10^{11} bits RAM	10^{11} neurons
	10^{15} bits disk	10^{13} bits disk	10^{14} synapses
Cycle time	10^{-9} sec	10^{-9} sec	10^{-3} sec
Operations/sec	10^{15}	10^{10}	10^{17}
Memory updates/sec	10^{14}	10^{10}	10^{14}

Figure 1.3 A crude comparison of the raw computational resources available to the IBM BLUE GENE supercomputer, a typical personal computer of 2008, and the human brain. The brain's numbers are essentially fixed, whereas the supercomputer's numbers have been increasing by a factor of 10 every 5 years or so, allowing it to achieve rough parity with the brain. The personal computer lags behind on all metrics except cycle time.

Answer:

The answer to the above question is as per the below

The sea slug Aplysia has approximately the same cycle period as a human brain = 10^3.

So the computational power in terms of memory updates per second = $\frac{20,000}{10^3} = 2 \times 10^7$ memory updates per second.

The high-end computer computational power is 10^{14} memory updates per second, so a high-end computer is twice as fast.

1.5 To what extent are the following computer systems instances of artificial intelligence:

• Supermarket bar code scanners.

Function: Simply scans a barcode and retrieves a price from a database.
AI Involvement: None or minimal. The process is entirely **rule-based** with no learning or reasoning involved.
Conclusion: Not an AI system, just a simple **automated lookup system**.

•Web search engines.

Function: Search engines (like Google or Bing) retrieve relevant web pages based on user queries.
AI Involvement:

- Use **natural language processing (NLP)** to understand queries.
- Implement **machine learning** to rank search results based on relevance.
- Employ AI techniques like **PageRank, semantic search, and personalization**.

Conclusion: Definitely AI-powered, using sophisticated **machine learning and NLP** techniques.

• Voice-activated telephone menus.

Function: Interprets spoken commands and directs callers to the correct department.
AI Involvement:

- Uses **speech recognition** (AI-driven).
- Often follows **predefined rules** rather than learning from experience.
- Limited ability to **understand context** or engage in **natural conversations**.

Conclusion: Basic AI, but not highly intelligent—relies on predefined responses.

• Spelling and grammar correction features in word processing programs.

Function: Detects and corrects spelling/grammar mistakes in documents.
AI Involvement:

- Older versions relied on **static rule-based approaches**.
- Modern tools (like Grammarly) use **machine learning, deep learning, and NLP** to detect **contextual** errors.
- Can suggest style improvements and learn from user preferences.

Conclusion: AI-powered, especially modern grammar checkers using NLP.

• Internet routing algorithms that respond dynamically to the state of the network.

Function: Dynamically adjusts network traffic routes based on congestion, failures, and efficiency.
AI Involvement:

- Uses **adaptive algorithms** that optimize performance.
- Employs **heuristic and reinforcement learning techniques** in some advanced systems.
- Does **not learn like a neural network**, but can adapt over time.

Conclusion: AI-like but not true AI—more **complex rule-based optimization** than deep learning.

System	AI Involvement?	Notes
Supermarket Bar Code Scanners	❌ No AI	Simple lookup system.
Web Search Engines	✅ Strong AI	Uses NLP & machine learning.
Voice-Activated Menus	⚠️ Weak AI	Limited speech recognition.
Grammar Checkers	✅ AI-Powered	Uses NLP & machine learning.
Internet Routing Algorithms	⚠️ AI-Like	Adaptive but not truly AI.

1.6 Is AI a science, or is it engineering? Or neither or both? Explain.

➢ **AI as a Science**

AI is a **scientific discipline** because it seeks to understand and replicate intelligence, often by studying cognitive processes, learning, reasoning, and problem-solving. Researchers in AI ask fundamental scientific questions, such as:

- How can machines learn from data?
- What are the mathematical foundations of intelligence?
- Can we model human reasoning in a formal way?

Fields like **machine learning theory, neural networks, natural language processing (NLP), and robotics** involve significant scientific exploration, mathematical modeling, and experimentation. AI also overlaps with **cognitive science, neuroscience, psychology, and philosophy**, further emphasizing its role as a scientific pursuit.

➢ **AI as Engineering**

AI is also an **engineering discipline** because it involves designing and building systems that perform intelligent tasks. Engineers develop practical AI applications such as:

- Self-driving cars
- Medical diagnosis systems
- AI chatbots and virtual assistants
- Fraud detection systems

AI engineers take the scientific principles discovered by researchers and turn them into working systems, using software engineering, data science, and optimization techniques.

➢ **Is AI Neither Science nor Engineering?**

Some might argue that AI is neither purely science nor purely engineering because:

- It does not always follow the traditional **scientific method** (it often relies on empirical results rather than formal proofs).
- It is not purely **engineering**, as AI systems do not always have predictable, deterministic behavior like traditional engineering systems.
- AI includes elements of **art, creativity, and philosophy**, making it a unique discipline.

➢ **Is AI Both Science and Engineering? (Yes!)**

In reality, AI is a hybrid field, requiring both:

- **Scientific exploration** to develop theories of intelligence and learning.
- **Engineering expertise** to build real-world applications that use these theories.

✓ **Conclusion**

AI is **both a science and an engineering discipline**—it seeks to understand intelligence scientifically while also creating practical systems that exhibit intelligent behavior. This dual nature makes AI one of the most exciting and interdisciplinary fields in modern technology.

1.7 "Surely animals cannot be intelligent—they can do only what their genes tell them."
Is the latter statement true, and does it imply the former?

➢ **Are Animals Only Controlled by Their Genes? (No)**

- While genes **influence** behavior, they do not rigidly determine everything an animal does.
- Many animals exhibit **learning, problem-solving, and adaptation**, which go beyond mere genetic programming.
- **Example:**
    - Birds like crows and ravens use **tools** to obtain food, a skill that is learned rather than purely instinctive.
    - Primates, dolphins, and elephants show **self-awareness, cooperation, and social learning**—all of which involve intelligence.
    - Even simpler animals, like octopuses, demonstrate **curiosity and problem-solving** abilities.

➢ **Does Genetic Influence Mean No Intelligence? (No)**

- Intelligence does not require complete independence from genetic programming.
- **Humans, too, are influenced by genes**, yet we are considered intelligent.

- Learning, memory, and decision-making are part of **biological intelligence**, regardless of genetic influence.

➢ What is Intelligence?

- If we define intelligence as **the ability to learn, adapt, and solve problems**, then many animals clearly qualify.
- Intelligence is often measured by:
  - **Problem-solving ability** (e.g., tool use in primates).
  - **Social intelligence** (e.g., cooperative hunting in wolves).
  - **Learning from experience** (e.g., rats navigating mazes).

Conclusion

- The claim that animals act *only* based on genetic programming is **false**—many animals learn and adapt.
- Even if genes play a role, this **does not disqualify them from being intelligent**.
- **Intelligence exists on a spectrum**, and many animals demonstrate forms of intelligence, even if it differs from human intelligence.

1.8 "Surely animals, humans, and computers cannot be intelligent—they can do only what their constituent atoms are told to do by the laws of physics." Is the latter statement true, and does it imply the former?

➢ Is the Statement About Physics True?

Yes, in a **strict physical sense**, everything—humans, animals, and computers—obeys the fundamental **laws of physics**.

- Every thought, action, and decision is ultimately a result of **physical processes** at the atomic and molecular level.
- In humans and animals, **neurons fire** based on electrical and chemical signals.
- In computers, transistors operate according to the laws of **electromagnetism and quantum mechanics**.

However, just because intelligence **emerges from physical interactions** does not mean it is **meaningless or nonexistent**.

➢ Does This Mean Intelligence Doesn't Exist? (No!)

Even if intelligence is **physically determined**, that does not mean it is **not real**.

- **Emergent properties:** Many complex phenomena (such as consciousness, intelligence, and emotions) **emerge** from simple physical laws, just as life emerges from basic chemistry.
- **Example:**

- Water molecules obey physics, but "wetness" is an **emergent property** that cannot be explained just by looking at a single water molecule.
- Similarly, intelligence arises from neurons or transistors, but it is **more than just** the physics of individual components.

➢ **Intelligence is a Higher-Level Concept**

- Intelligence is **not about individual atoms** but about **how they interact in complex ways** to process information, make decisions, and solve problems.
- Whether in **humans, animals, or AI**, intelligence is a **real phenomenon**, even if it is physically determined.
- Saying "intelligence doesn't exist because everything follows physics" is like saying "storms don't exist because they are just moving air molecules." Clearly, storms exist!

➢ **Does Physics Limit Intelligence?**

Not necessarily!

- **Deterministic Physics?** Some interpretations of physics suggest that everything is predetermined, but **quantum mechanics introduces uncertainty** at a fundamental level.
- Even if intelligence is fully **deterministic**, it does not mean it is an illusion—it simply means it follows physical laws, just like everything else.

Conclusion

- The **laws of physics do govern everything**, including intelligence.
- However, this **does not mean intelligence does not exist**—it simply arises from **complex interactions** within those physical laws.
- **Humans, animals, and AI can still be intelligent**, even if their intelligence is ultimately the result of physical processes.

1.9 There are well-known classes of problems that are intractably difficult for computers, and other classes that are provably undecidable. Does this mean that AI is impossible?

No, the existence of intractably difficult or undecidable problems does not mean that AI is impossible. These limitations apply to all computational systems, including AI, but they do not prevent AI from being useful, powerful, or even achieving human-like intelligence.

Here's why:

1. **Not All Problems Are Intractable**
   Many problems AI needs to solve are **tractable**—that is, they can be computed efficiently within practical time limits. AI systems are already highly effective at tasks like image recognition, natural language processing, and strategic game playing because these problems fall within classes that are computationally feasible.

2. **Approximation and Heuristics**
   Even for difficult problems (such as NP-hard problems), AI can use **approximation algorithms, heuristics, and machine learning** to find good-enough solutions in a reasonable time. AI doesn't always need to find the absolute best answer—just a useful one.
3. **Undecidable Problems Don't Block AI**
   Some problems are **provably undecidable** (e.g., the Halting Problem), meaning no algorithm can always provide a correct solution. However, AI doesn't need to solve undecidable problems to be effective. Humans also operate under similar limitations but still function intelligently.
4. **AI as a Practical Engineering Field**
   AI is about **practical intelligence, not theoretical perfection**. While some problems remain intractable, AI systems can still navigate the real world effectively by using probabilistic reasoning, learning from data, and computational power.
5. **Human Intelligence Also Has Limits**
   Human intelligence itself does not solve all problems perfectly or instantly. Humans rely on approximations, intuition, and experience—just like AI. Theoretical limits of computation do not necessarily limit AI's ability to perform useful tasks.

In summary, AI is possible and already successful despite computational limitations. The existence of hard or undecidable problems does not mean that useful intelligence, problem-solving, or learning cannot be achieved.

1.10 Why would evolution tend to result in systems that act rationally? What goals are such systems designed to achieve?

Evolution tends to result in systems that act rationally because rational behavior often leads to better survival and reproduction outcomes. Organisms that can make rational decisions, such as assessing risks, finding resources efficiently, and avoiding danger, are more likely to thrive and pass on their genes to future generations. The goals of such systems designed by evolution are primarily aimed at enhancing the organism's chances of survival and reproduction within its environment. These goals include finding food, shelter, mates, avoiding predators, and successfully raising offspring. Essentially, the systems are designed to maximize the organism's fitness, which is its ability to survive and reproduce in its specific ecological niche.

1.11 Examine the AI literature to discover whether the following tasks can currently be solved by computers:

1. Playing a decent game of table tennis (Ping-Pong): Yes, there are AI systems that can play table tennis at a decent level. For example, there are AI-powered robots designed specifically for playing table tennis that can compete against human opponents.
2. Driving in the center of Algiers, Algeria: While self-driving technology has advanced significantly, driving in complex urban environments like Algiers remains challenging due to factors like chaotic traffic patterns, unpredictable pedestrian behavior, and lack of standardized road infrastructure. Progress is being made, but fully autonomous driving in such extreme environments is not yet widespread.
3. Driving in San Francisco, California: Yes, currently one of the few cities in the world where the public can hail a an autonomous driverless robo-taxi.(Tesla, Waymo)

4. Buying a week's worth of groceries at the market: No. No robot can currently put together the tasks of moving in a crowded environment, using vision to identify a wide variety of objects, and grasping the objects (including squishable vegetables) without damaging them. The component pieces are nearly able to handle the individual tasks, but it would take a major integration effort to put it all together.
5. Buying a week's worth of groceries on the Web: Yes, Software robots are capable of handling such tasks, particularly if the design of the web grocery shopping site does not change radically over time.
6. Discovering and proving new mathematical theorems: Yes. For example, the proof of Robbins algebra described on page 360. The computer proof of the Robbins Problem was completed on October 10, 1996 by a program named "EQP".
7. Writing an intentionally funny story: AI systems can generate text that can be humorous, but writing a genuinely funny story that resonates with human audiences in a nuanced way remains a challenging task. AI-generated humor often lacks the subtlety, context awareness, and understanding of cultural nuances that humans possess.
8. Translating spoken English into spoken French in real time: Real-time speech translation systems exist, and some can translate between English and French with decent accuracy. While they may not be perfect, they can facilitate basic communication in real-time scenarios.
9. Performing a complex surgical operation: Yes. Robots are increasingly being used for surgery, although always under the command of a doctor. Robotic skills demonstrated at superhuman levels include drilling holes in bone to insert artificial joints, suturing, and knot-tying. They are not yet capable of planning and carrying out a complex operation autonomously from start to finish.